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Cover Letter

Joshua Behymer

Deaf and Hard of Hearing Accommodation Specialist

Disability Services for Students

Western Michigan University

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06/16/2025

Kalamazoo Community Foundation

402 E Michigan Ave, Kalamazoo, MI 49007

Dear Review Committee,

On behalf of Disability Services for Students (DSS) at Western Michigan University (WMU), I am pleased to submit this proposal requesting a one-time grant of $30,000 to support accessibility infrastructure improvements for BroncoNav, our campus paratransit shuttle initiative. BroncoNav addresses transportation equity for students with disabilities by offering on-demand mobility support and is aligned with the Kalamazoo Community Foundation's (KCF) commitment to educational equity, youth development, and inclusive communities. The BroncoNav fits in with the Education portion of KCF’s WHEN goals, specifically “Removing barriers to education by addressing basic needs like transportation and technology.”

The DSS office has committed to staffing and maintaining the program, and we are seeking grant support for one-time infrastructure and technology purchases critical to successful implementation. These include accessibility signage, a custom rider app, and advertising-enabled features that will ensure the long-term sustainability of the program.

With your support, BroncoNav will improve daily access to classes, reduce disability-related absenteeism, and contribute to retention and well-being among WMU students.

Thank you for considering this proposal. I would be honored to provide additional information or meet to discuss BroncoNav and its impact.

Sincerely,

Joshua Behymer
Deaf and Hard of Hearing Accommodation Specialist
Disability Services for Students, WMU

Executive Summary

BroncoNav is an innovative paratransit shuttle program serving students registered with Disability Services for Students (DSS) at Western Michigan University. It provides timely, reliable, and inclusive transportation across WMU's expansive and hilly campus. The service addresses a critical equity gap for students with mobility and mental health challenges who face barriers to academic participation due to transportation issues.

DSS is seeking a one-time $30,000 grant from the Kalamazoo Community Foundation to fund key infrastructure and technological enhancements including GPS vehicle tracking, LED signage, a student-facing app, survey kiosks, and advertising panels to support long-term sustainability. Staff positions have already been funded by WMU.

BroncoNav aligns with the Foundation's goals by improving access to education, supporting youth well-being, and strengthening equitable campus infrastructure. Through a robust evaluation plan, WMU will monitor outcomes including ridership, user satisfaction, academic engagement, and cost efficiency.

This investment will enhance access, retention, and inclusion for hundreds of WMU students, creating a scalable model for other institutions.

Statement of Need

A significant number of WMU students with disabilities experience transportation barriers that prevent consistent class attendance and engagement. Existing public transit systems do not adequately serve students with physical, cognitive, or temporary mobility limitations. In fact, WMU is prioritizing having a more walkable campus per the WMU 2025 Master Plan. The

downside to a walkable campus is that as fixed bus routes and parking lots get further out from the heart of campus, that inherently makes getting there more difficult for those with both visible and invisible disabilities. The campus's steep terrain, inclement weather, and insufficient parking exacerbate access challenges. This is why the BroncoNav is available to all registered DSS students – no matter their documented diagnosis.

Research confirms that lack of reliable transportation directly impacts academic performance and retention. National data shows that students with transportation difficulties are 36% more likely to miss classes and 43% more likely to consider dropping out (Trellis Research Services, 2023). Programs like BroncoNav reduce these risks by offering adaptive, user-focused transportation.

Universities such as the University of Michigan, Ohio State University, and Arizona State University have implemented similar campus-based paratransit systems with notable results in increased student satisfaction and retention (Urban Institute, 2023; Georgetown Civic Mapping Initiative, 2022). These examples show a direct correlation between reliable campus transit access and improved educational outcomes, making BroncoNav a crucial investment for WMU.

The lack of an accessible, on-demand campus shuttle places WMU students at a disadvantage compared to peers at similar institutions. BroncoNav will fill this critical gap, enabling timely attendance, improving mental health, and supporting WMU’s broader diversity, equity, and inclusion mission.

Project Description

BroncoNav is a campus-based paratransit shuttle initiative designed to improve mobility, equity, and academic engagement for students registered with Disability Services for Students (DSS) at Western Michigan University (WMU). Developed in partnership with Revision Autonomy—an engineering startup launched by WMU alumni—BroncoNav merges inclusive design with innovative transportation technology to address longstanding accessibility gaps across campus.

Overview of Services

BroncoNav will deliver on-demand, door-to-door transportation for DSS-registered students navigating WMU's expansive and topographically challenging campus. The service will operate using an electric shuttle vehicle that is fully wheelchair accessible and equipped with advanced navigation and GPS tracking systems. Key features of the service include:

- **Accessible Electric Shuttle Van**: Designed with safety and usability in mind, the shuttle includes a wheelchair lift, safety restraints, and real-time GPS tracking to monitor route performance and improve user experience.
- **Mobile App Integration**: A user-facing application will allow riders to schedule rides, view shuttle arrival times, receive alerts, and provide service feedback. The app will be ADA-compliant, with high-contrast, screen-reader-friendly design.
- **Wayfinding and Signage**: Ten custom-designed outdoor sign stands will mark designated pick-up and drop-off zones across campus, providing clear visibility for students and promoting program awareness.
- **Real-Time Feedback Tools**: Riders can complete brief surveys using QR codes posted inside the shuttle and at stops. Feedback will be compiled and analyzed regularly to guide improvements.
- **Advertising-Enabled LED Panel**: A shuttle-mounted digital advertising panel will support long-term program sustainability by generating revenue from sponsors aligned with WMU's accessibility and equity values.

Implementation Timeline

The BroncoNav project will be implemented in four distinct phases from May 2025 through July 2027:

- **Summer 2025 (Phase 1 – Planning and Setup)**:
    - Form project team and clarify responsibilities among DSS, Revision Autonomy, and other WMU partners.
    - Launch campus-wide survey to identify high-need transit zones and preferred service hours.
    - Procure equipment and begin installing signage and shuttle hardware.
- **Fall 2025 (Phase 2 – Service Launch)**:
    - Conduct shuttle driver training and DSS student orientations.
    - Deploy the BroncoNav mobile app and begin full-service operations.
    - Launch QR-coded surveys and feedback collection efforts.
- **Summer 2026 (Phase 3 – Midpoint Evaluation)**:
    - Analyze ridership, user feedback, and GPS data to assess service effectiveness.
    - Adjust routes, schedules, and app functionality based on student input.
    - Release a mid-cycle evaluation report and prepare for autonomous pilot testing.
- **Fall 2026 to Summer 2027 (Phase 4 – Pilot and Evaluation)**:
    - Introduce autonomous vehicle functionalities under supervision, gathering technical and experiential data.
    - Continue tracking impact on attendance, satisfaction, and academic performance.
    - Publish final evaluation report by July 2027 to summarize outcomes, lessons learned, and potential for broader replication. (See Appendix A)

Program Objectives

BroncoNav aims to directly improve academic persistence and campus participation among DSS-registered students by:

- Providing equitable transportation that reduces physical and logistical barriers to campus engagement.
- Increasing class attendance and reducing the frequency of missed assignments due to transportation issues.
- Promoting autonomy and independence for students with mobility or mental health challenges.
- Enabling DSS and WMU to monitor and improve services through real-time data analysis and stakeholder input.

In addition, BroncoNav will serve as a model for scalable, student-centered accessibility programs in higher education. Lessons learned through implementation will be shared with peer institutions through DSS networks and academic conferences.

Organizational Roles and Partnerships

BroncoNav is led by the DSS Office at WMU, a department with over three decades of experience supporting students with disabilities. DSS staff will manage project coordination, rider support, and data collection. Joshua Behymer, Accommodation Specialist and project lead, brings both lived experience and professional expertise in equity-centered service design.

Revision Autonomy will lead technical implementation, including GPS tracking, data dashboard development, and the integration of autonomous shuttle features. Co-founder Dr. Zachary Asher will oversee all hardware and software elements, as well as the safe implementation of autonomous driving trials.

Additional collaborators include:

- **WMU Office of Diversity and Inclusion (ODI)**: Oversight and evaluation support.
- **WMU Facilities Management**: Coordination of signage placement and campus infrastructure.
- **Student Interns**: Support for communications, evaluation, and shuttle logistics.
- **Finance Office**: Budget tracking and fiscal reporting in line with KCF grant guidelines.

Innovation and Sustainability

BroncoNav distinguishes itself from traditional paratransit services through its use of real-time technology and commitment to sustainability. The inclusion of a revenue-generating LED panel and the development of a custom mobile app positions the program for long-term viability beyond the grant period.

The integration of autonomous technology also future-proofs the service. While initial operations will include a human driver, BroncoNav's phased plan allows for the careful, student-informed introduction of driverless features—an innovation that could substantially reduce operating costs and scale access over time.

Broader Impact

The benefits of BroncoNav will extend beyond the students it serves directly. By enhancing visibility and understanding of mobility-related barriers on campus, the program reinforces WMU's institutional commitment to diversity, equity, and inclusion. It also aligns with emerging best practices in accessible campus design and sustainable transportation.

In doing so, BroncoNav supports both the academic success of marginalized students and the overall vibrancy of campus life—goals shared by the Kalamazoo Community Foundation and embedded in its WHEN framework.

Evaluation Plan

BroncoNav will use a Utilization-Focused Outcome Evaluation framework with a goal of measuring effectiveness, accessibility, and impact. (See Appendix B)

Key Indicators:

- Total number of rides and unique riders (monthly logs)
- On-time arrival performance (real-time GPS data)
- Rider satisfaction (daily QR surveys)
- Perceived accessibility and academic impact (annual interviews)
- Budget tracking and cost-efficiency (quarterly reports)

Methods:

- Mixed methods combining digital data collection, surveys, interviews, and focus groups
- Data analysis using Excel and thematic coding
- Dashboards and reports for internal and funder review

Timeline:

- Summer 2025: Baseline data
- Fall 2025–Spring 2027: Ongoing monitoring and user feedback
- Summer 2026: Mid-cycle evaluation
- Summer 2027: Final report with outcome summary

Evaluation will be conducted by DSS staff in partnership with Revision Autonomy.

Budget & Budget Justification

Category	Description	Cost
App Development	Development and enrollment (Apple/Google)	$4,624
GPS Tracking & Notifications	Hardware, API, messaging tools	$2,450
Admin Tools	Dashboard and backend integration	$1,300
Signage & Advertising	Outdoor signs and shuttle-mounted LED panel	$15,000
Indirect Costs	10% to WMU, 3% to ODI	$4,000
Contingencies	Unexpected costs	$1,126
Total		$30,000

The proposed budget totals **$30,000** and is structured to support the successful implementation and long-term sustainability of BroncoNav. All expenditures are one-time costs focused on infrastructure, technology, and administrative tools. No personnel costs are requested, as these are fully covered by institutional resources.

App Development ($4,624)

The mobile application is a critical component of BroncoNav, enabling students to schedule rides, receive notifications, and provide service feedback. This line includes:

- Enrollment in Apple and Google developer accounts ($99 and $25, respectively).
- Development costs for a student-facing mobile application, fully compliant with accessibility standards.
- Integration of in-app communication tools for real-time messaging and support.

The app ensures accessible, user-centered interaction with the shuttle system, meeting the diverse needs of students with mobility and cognitive challenges.

GPS Tracking and Notifications ($2,450)

Real-time tracking is essential for operational efficiency and user satisfaction. This line includes:

- GPS hardware and installation for the shuttle vehicle.
- Twelve months of GPS application programming interface (API) access and live location tracking.
- Integration with Twilio for SMS notifications to riders and automated system alerts.

These tools support both service delivery and evaluation by capturing ride data, wait times, and route performance.

Administrative Tools ($1,300)

An administrative dashboard will allow DSS staff to monitor usage metrics, review rider feedback, and adjust schedules as needed. These tools are essential for real-time service management and long-term performance evaluation.

Signage and Advertising ($15,000)

This category supports both physical infrastructure and sustainability planning:

- Ten custom outdoor signs to identify shuttle pick-up and drop-off zones.
- A digital LED display panel mounted on the shuttle to promote campus resources and, eventually, accessibility-aligned sponsorships.

The LED panel represents a long-term investment in financial sustainability, allowing BroncoNav to generate revenue beyond the grant period through advertising partnerships.

Indirect Costs ($4,000)

In accordance with WMU's institutional policy, indirect costs are calculated at 10% of the total award for facilities and administrative support ($3,000) and 3% for support from the Office of Diversity and Inclusion ($1,000). These costs ensure appropriate oversight, reporting, and infrastructure access.

Contingency ($1,126)

A contingency fund is included to account for unexpected technical or procurement-related expenses. As this is a new initiative, a contingency is prudent to mitigate risk without requiring additional funding.

Total Requested: $30,000

This proposed budget aligns directly with the project's objectives to improve access, reduce barriers, and support long-term program sustainability. Each line item reflects a thoughtful investment in infrastructure, evaluation capacity, and inclusive technology, consistent with WMU's mission and the goals of the Kalamazoo Community Foundation. (See Appendix C)

Organizational Information/Capacity

Disability Services for Students (DSS) at WMU has over 30 years of experience providing academic accommodations and supporting students with disabilities. The department is committed to accessible education and currently serves well over one thousand students each semester. (See Appendix D)

BroncoNav project is led by Joshua Behymer, Accommodation Specialist, in conjunction with Dr. Zachary Asher of Revision Autonomy. The team combines administrative, technical, and academic expertise with lived experience and established student partnerships. DSS has internal data collection systems, strong relationships with students and departments, and the operational infrastructure needed to implement and sustain the service. DSS has access to students served and will be tasked with the collection of qualitative data including from surveys and focus groups. Dr. Asher with Revision Autonomy and the Energy Efficient Autonomous

Vehicle Lab will be responsible for collecting quantitative data such as raw ridership numbers, ride timeliness, and number of registered students at Bronconav.com website.

Joshua Behymer. As an Accommodation Specialist, Joshua works daily with students and faculty to address academic, housing, and campus-living needs. Each semester, Joshua sees how students are impacted through lack of paratransit. Whether the lack of engagement comes from an academic or social perspective, we know that all of it can contribute to poor physical and mental health, leading to absenteeism and stopping-out. Joshua has written over a dozen letters of support for Hardship Withdrawals and assisted many other students negotiate incompletes for plenty of others. Before coming to work at the DSS office at WMU, Joshua spent almost a decade in Supported Employment, working with those living with disabilities to assist in obtaining – but more importantly – maintaining employment in the community. He brings these past experiences to the project as an advocate, communicator, and connector, which should serve the project well.

Dr. Zachary Asher. An accomplished PhD in Mechanical and Aerospace engineering, Dr. Asher is very familiar with quantitative data analysis and giving deliverables. Dr. Asher is also a co-founder of Revision Autonomy, a start-up created by, and for WMU faculty and alumni. Dr. Asher boasts over $4 million in awarded grants in the energy efficient and autonomous vehicle spaces. He will prove as an invaluable resource for the BroncoNav when it comes to data collection and analysis as well as generating deliverables for the grant funder.

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Appendices

Appendix A: Timeline

Task	Start Date	End Date
Planning & Team Formation	May 2025	June 2025
Survey & Community Feedback	June 2025	August 2025
Infrastructure Setup	July 2025	September 2025
Staff & Rider Training	August 2025	September 2025
Service Launch	September 2025	Ongoing
Mid-Cycle Evaluation	May 2026	July 2026
Autonomous Pilot	September 2026	April 2027
Final Evaluation	May 2027	July 2027

Appendix B: Evaluation Plan Summary Table

Indicator	Data Source	Method	Frequency	Collector
Number of users served	Ride logs	Digital records	Monthly	Revision Autonomy
Average lead time	GPS	Automated tracking	Monthly	Revision Autonomy
On-time ride performance	GPS/timestamps	Real-time system	Monthly	Revision Autonomy
Rider satisfaction	Rider surveys	QR/text surveys	Daily	DSS Office
Accessibility perception	Focus groups	Group sessions	Annual	DSS Office
Academic participation	Student interviews	Semi-structured	Annual	DSS Office
Operational cost	Budget records	Financial analysis	Quarterly	DSS/ODI Finance
Stakeholder satisfaction	Staff surveys	Digital/printed	Annual	DSS Office

Appendix C: Budget Estimate Table

Category	Item Description	Quantity	Unit Cost	Total Cost
App Development	App Store Enrollment (Apple)	1	$99.00	$99.00
	App Store Enrollment (Google)	1	$25.00	$25.00
	Mobile App Dev Time	1	$4,500.00	$4,500.00
GPS Tracking	Live Vehicle Tracking Setup	1	$1,500.00	$1,500.00
	Monthly GPS API Usage (est.)	12	$20.00	$240.00
Notifications	Twilio SMS Integration	1	$500.00	$500.00
	SMS Messaging (est. per year)	12	$7.50	$90.00
	Real-time Messaging	12	$10.00	$120.00
	In-app Messaging System	1	$1,500.00	$1,500.00
Admin Tools	Admin Dashboard	1	$1,300.00	$1,300.00
Advertising	Custom, outdoor signs at pickup/drop-off locations	10	$500.00	$5,000.00
	LED ad screen – shuttle mountable	1	$10,000.00	$10,000.00
Indirect Costs	10% of total award to WMU	1	$3,000.00	$3,000.00
	3% of total award to ODI	1	$3,000.00	$1,000.00
Contingencies	Overages	1	$1,126.00	$1,126.00
	Total			$30,000.00

Appendix D: Stakeholder Roles Overview

Stakeholder	Role/Responsibility
DSS Project Lead	Overall coordination, stakeholder communication
Dr. Zachary Asher	Technical implementation, data tracking
DSS Staff	Rider support, data collection
Revision Autonomy	GPS, dashboard tools, technology integration
Communication Officer	Draft reports, manage feedback collection tools
Riders (Students)	Daily use and feedback contributors
Finance Office	Budget tracking, expenditure monitoring

Appendix E: Logic Model

